# CS543 • WEB SERVICES

## ASSIGNMENT 6 OF 12

# Honest Responses: Status Codes, Errors & Validation

Group ID: 17

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# Part C — Validation and Content Formats

Validate AlgoViz input before algorithm execution

## C1. Short theory

Validation must happen before running an algorithm or saving history. Collect all field errors and return one 422 response. JSON is the default response format; XML may be returned only if the service implements it. If the requested Accept format cannot be produced, return 406 Not Acceptable.

## C6. Validation summary

| Rule | Expected handling |
|---|---|
| Missing algorithm | 422 with algorithm field error. |
| Unsupported algorithm | 422 with supported-algorithm error. |
| Empty input | 422 with input field error. |
| Input over configured limit | 422 with input-size error. |
| Malformed JSON | 400 Bad Request. |
| Unsupported Accept type | 406 Not Acceptable. |

## C2. Request validation DTO

```
public record VisualizationRequest(
@NotBlank String algorithm,
@NotEmpty List<@NotNull Integer> input) {}
```

## C3. Validate algorithm and input before processing

```
private static final Set<String> SUPPORTED = Set.of(
"BUBBLE_SORT", "SELECTION_SORT", "INSERTION_SORT",
"LINEAR_SEARCH", "BINARY_SEARCH");

public void validateRequest(VisualizationRequest req) {
List<String> errors = new ArrayList<>();

if (req.algorithm() == null ||
!SUPPORTED.contains(req.algorithm())) {
errors.add("algorithm must be supported");
}
if (req.input() == null || req.input().isEmpty()) {
errors.add("input must not be empty");
}
if (req.input() != null && req.input().size() > 500) {
errors.add("input must contain at most 500 values");
}
if (!errors.isEmpty()) {
throw new InvalidVisualizationInputException(errors);
}
// Only call the Java algorithm layer after validation passes.
}
```

## C4. Content-Type and Accept

```
// JSON response example
return ResponseEntity.ok()
.contentType(MediaType.APPLICATION_JSON)
.body(response);

// Problem Details errors should use:
// application/problem+json; charset=utf-8

// If XML is not implemented and the client requires XML,
// return 406 Not Acceptable rather than sending JSON as XML.
```

## C5. XML negotiation option

```
// If XML support is implemented, configure an XML message
// converter and a response representation for the endpoint.
// Then test with:
// curl -i -H "Accept: application/xml" \
// http://localhost:8080/api/algorithms
//
// Do not claim XML support until this test returns XML.
```

## Notes / implementation caution

The exact supported algorithms and maximum input length should match the current AlgoViz implementation. The sample limit of 500 is an illustrative project rule.